

Economic History II

The trade in enslaved Africans and its long-run consequences

FGV EPGE

The puzzle of African underdevelopment

- Africa's economic performance in the second half of the twentieth century: poor.
- The informal explanation: a history of extraction, in two events, the slave trades and colonialism.
- Official colonial rule, 1885 to about 1960: roughly 75 years.
- The slave trades, 1400 to 1900: nearly **500 years**.
- Empirical work had studied colonial rule. The slave trades, not at all.

“Slavery was corruption: it involved theft, bribery, and exercise of brute force as well as ruses. Slavery thus may be seen as one source of precolonial origins for modern corruption.”

Patrick Manning (1990, p. 124), quoted by Nunn

What Nunn does

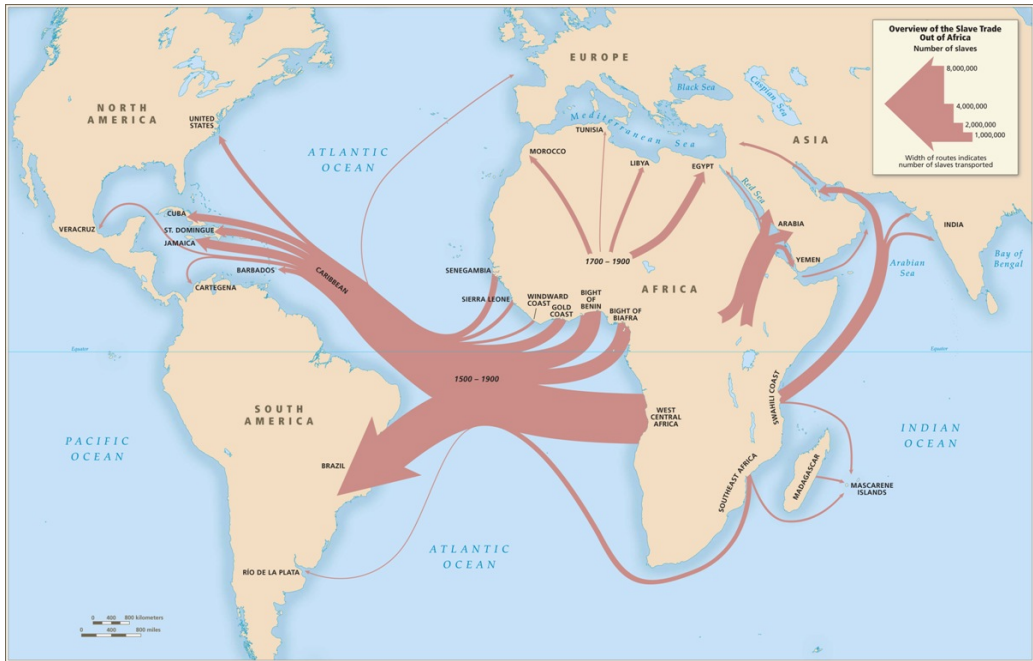
- First empirical examination of the slave trades and later economic development.
- Builds estimates of the number of slaves exported from **each African country** in each century, 1400 to 1900.
- Question: can part of Africa's current underdevelopment be explained by its slave trades?

Four slave trades

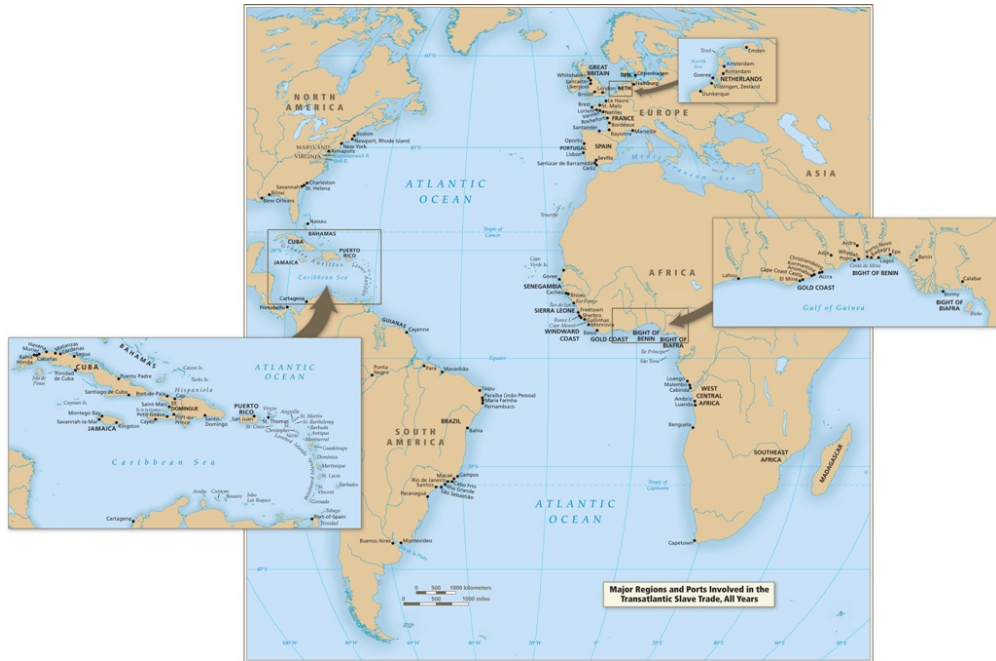
- Between 1400 and 1900 the continent experienced **four simultaneous** slave trades.
- Transatlantic, the largest and best known: from West Africa, West-Central Africa and Eastern Africa to the European colonies in the New World.
- Trans-Saharan: from south of the Sahara to Northern Africa.
- Red Sea: from inland of the Red Sea to the Middle East and India.
- Indian Ocean: from Eastern Africa to the Middle East, India, or plantation islands.

The volume

- Transatlantic trade alone: approximately **12 million** slaves exported. Another 6 million in the other three. [link](#)
- Not included in those figures: those killed during the raids, and those who died on the journey to the coast.
- Mortality crossing the Atlantic: estimates of 7 to 20 percent. On the trek to the coast: 10 to 50 percent.
- Manning's calculation: by 1850 Africa's population was only *half* of what it would have been without the slave trades.



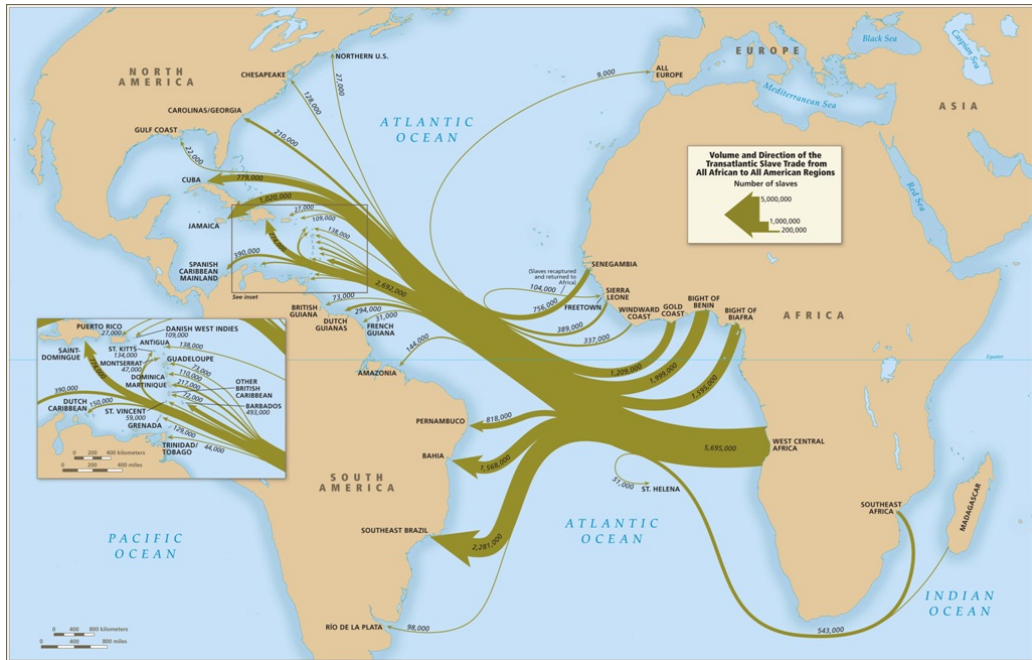












Enslavement and fragmentation

- Unlike previous slave trades: individuals of the same or similar ethnicities enslaved one another.
- The most common manner of taking slaves: villages or states raiding one another.
- Where groups of villages had developed into village federations, relations between them turned hostile.
- Ties weakened, which impeded the formation of larger communities and broader ethnic identities.
- Kusimba: *“insecurity confined people within ethnic boundaries constructing spheres of interaction.”*

The gun-slave cycle

- An environment of uncertainty and insecurity. Individuals needed weapons: iron knives, spears, swords, later firearms.
- Weapons came from Europeans in exchange for slaves, often obtained through local kidnappings.
- More insecurity, more need to enslave others in order to protect oneself.
- Historians name the cycle: the **gun-slave cycle**, or the iron-slave cycle.
- Result: communities raided other communities, and members of a community raided within it. Documented among the Balanta of Guinea-Bissau, the Minyanka of Mali, the Makua, Chikunda and Yao of East Central Africa.

States that collapsed

- Internal conflict brought political instability and, in many cases, the collapse of preexisting forms of government.
- Sixteenth-century northern Senegambia: the Portuguese slave trade disintegrated the **Joloff Confederation**, replaced by the much smaller kingdoms of Waalo, Kajoor, Baol, Siin and Saalum.
- Southern Senegambia: complex state systems were evolving before the trade. The evolution stagnated after the Portuguese arrived.
- Kongo, West-Central Africa: as early as 1514 the kidnapping of local citizens for sale to the Portuguese had become rampant.

*“there are many traders in all corners of the country. They bring ruin to the country.
Every day people are enslaved and kidnapped, even nobles, even members of the
king’s own family”*

Affonso, king of Kongo, writing to Portugal in 1526 (Vansina 1966, p. 52)

What replaced the old states

- Preexisting governance structures gave way to small bands of slave raiders under a ruler or warlord.
- The bands generally could not develop into large, stable states.
- Colson: they retained “an air of improvisation”. Few band leaders could hand power to a legitimate successor.
- Leadership was a personal role rather than an established office.
- By the end of the slave trades, few ancient states remained among the Bantu-speaking ethnicities.

The corruption of justice

- The slave trades also corrupted previously established **legal structures**.
- It became common to obtain slaves by falsely accusing others of witchcraft or other crimes.
- Klein: *“communities began enslaving their own. Judicial penalties that formerly had taken the form of beatings, payment of compensation or exile, for example, were now converted to enslavement.”*
- Leaders themselves supported, and sometimes instigated, the abuse of the judicial system. Slaves obtained this way were paid as tribute to avoid being raided.

The red water ordeal

- The Cassanga of modern day Guinea-Bissau, documented by Hawthorne.
- The chief used the **red water ordeal** to procure slaves and their possessions.
- Those accused of a crime were forced to drink a poisonous red liquid.
- Vomiting was judged not guilty. Not vomiting usually brought death by poisoning.
- Possessions were then seized, and family members sold into slavery.

Measuring the slave trades, country by country

First source: shipping data

- Measure of interest: the total number of slaves taken from each country during the four slave trades, 1400 to 1900.
- Shipping data report the number of slaves exported from each port or region.
- Transatlantic: the Trans-Atlantic Slave Trade Database (Eltis et al. 1999), **34,584 voyages** from 1514 to 1866.
- Estimated to contain 82 percent of all transatlantic slaving voyages ever attempted.
- After 1700, 77 percent of voyages have shipping information from more than one source. Six sources on average.

Second source: slave ethnicities

- A port of departure does not say where a slave was captured. Many came from countries further inland.
- Records of sale, slave registers, runaway notices, court records, church records, notarial documents.
- A slave's “**nation**” was identified by name, and by ethnic markings: cuts, scars, hairstyles, the filing of teeth.
- Slaves were legally property, so buyers and sellers had a strong incentive to record the birthplace correctly.

How much material

- Transatlantic: 54 different samples, **80,656 slaves**, 229 distinct ethnic designations.
- Indian Ocean: six samples, 21,048 slaves, 80 ethnicities. Red Sea and trans-Saharan: two samples each, and far fewer slaves.
- Brazilian records are 12 of those 54 samples: Bahia, Salvador and Rio de Janeiro.
- The record types there: slave lists, manumission certificates, probate records, court records, prison records, death certificates.

Where the ethnicities come from: Murdock's map

- Mapping ethnicities into countries drew on past work by African historians, linguists and ethnographers: Koelle (1854), Murdock (1959), Curtin (1969), Higman (1984), Hall (2005).
- Both papers use the ethnographic classification of **George Peter Murdock (1959)**, which maps ethnic boundaries as of the late nineteenth century.
- Those boundaries predate the political borders Europeans drew afterwards.
- Ethnicities are much smaller than modern countries, so they generally map cleanly into one country.
- Where an ethnicity falls in more than one country, land area is used as the weight.



FIGURE II
Ethnic Boundaries Defined by Murdock (1959) and Modern Political Boundaries



From the coast to the interior

- With ethnicity data, one can calculate the ratio of slaves from a coastal country relative to the landlocked countries behind it.
- That ratio splits each port's departures among the countries the slaves actually came from.
- Figure 1 is an artificial map of the west coast of Africa. Each square is a country.

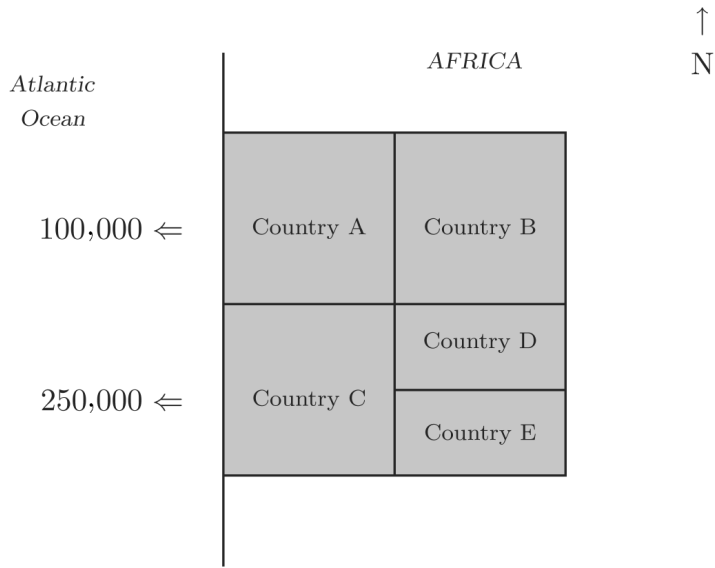


FIGURE I
An Artificial Map of the West Coast of Africa

Reading the artificial map

- Shipping data: 100,000 slaves shipped from Country A, 250,000 from Country C.
- Country B is landlocked behind A. Some of those shipped from A were from B.
- Ethnicity data give the ratio of A to B as 4 to 1. So 80,000 from A and 20,000 from B.
- C to D to E is 3 to 1 to 1: 150,000, 50,000 and 50,000.
- Done separately for each slave trade and for 1400 to 1599, 1600 to 1699, 1700 to 1799, 1800 to 1900.

Does the procedure work

- Three samples exist where both the ethnicity and the port of shipment are known.
- In those samples the procedure correctly identifies the origins of between **83 and 98 percent** of slaves.
- The country totals match what African historians already held: the “Slave Coast” (Benin and Nigeria), West-Central Africa (Angola and Congo), the “Gold Coast” (Ghana).
- Ethiopia and Sudan rank high through the Red Sea and Saharan trades. South Africa and Namibia exported “virtually no slaves”.
- Slaves from the deep interior are undercounted, because only those who survived the whole journey appear in the ethnicity samples.

The slave trades and income today

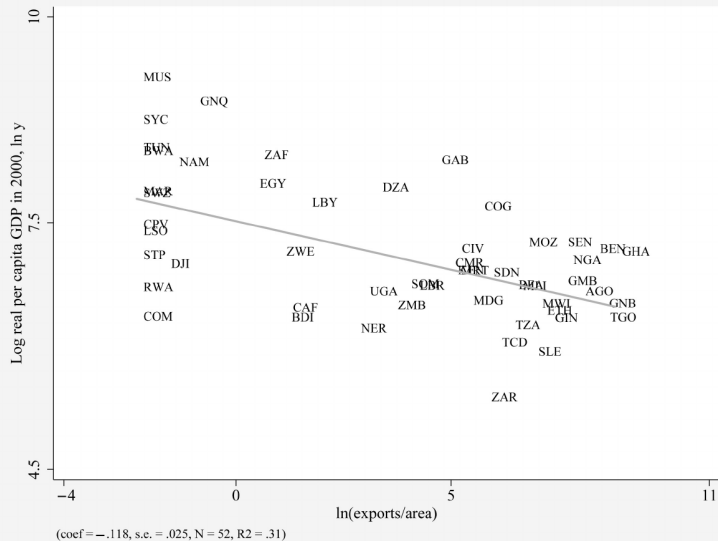


FIGURE III
 Relationship between Log Slave Exports Normalized by Land Area,
 $\ln(\text{exports}/\text{area})$, and Log Real Per Capita GDP in 2000, $\ln y$

Countries that lost the most are poorer today

- Slaves exported, normalized by land area, against per capita GDP in 2000 (Maddison 2003).
- The African countries that are the **poorest today** are the ones from which the most slaves were taken.
- The pattern survives accounting for the identity of the colonizer at independence.
- It survives geography and climate: distance from the equator, longitude, rainfall, humidity, temperature, proximity to the ocean.
- It survives accounting for gold, oil and diamond production.

How large

- Take a country at the mean African income of \$1,249.
- Move its slave export measure down by one standard deviation.
- Income rises to \$1,864, an increase of **50 percent**.
- Nunn presents this as illustrative, since it reads the simple relationship as if it were causal.

Did the slave trades cause it?

The selection problem

- An equally plausible alternative: the societies that were initially the most economically and socially **underdeveloped** selected into the slave trades, and remain underdeveloped today.
- Then the relationship would exist even if the slave trades had no effect at all.
- Nunn's first move: ask what kind of society actually selected into the slave trades.

Early trade required developed institutions

- Much of the early trade between Africans and Europeans was in commodities other than slaves.
- Only societies with sufficiently developed institutions could support that trade.
- Between 1472 and 1483 the Portuguese sailed south along the coast of West-Central Africa looking for trading partners, and found none north of the Zaire river.
- Vansina: *“the local coastal societies were just too small in terms of people and territory; their economic and social institutions were too undifferentiated to facilitate foreign trade.”*
- Sustained trade began with the **Kongo kingdom**: centralized government, a national currency, developed markets and trading networks.

The most prosperous were the most affected

- When European demand turned almost exclusively to slaves, the preference for the most developed parts of Africa continued.
- The more prosperous areas were also the most densely populated: large numbers of slaves could be obtained efficiently once conflicts were instigated.
- The most violent and hostile societies, and therefore the least developed, were often best able to resist. The trade in Gabon stayed limited for centuries because of the defiance of its inhabitants.
- Measured by **population density in 1400**, the parts of Africa most affected by the slave trades were the ones that had been most prosperous.
- Selection therefore pushes the simple comparison toward zero. If it errs, it understates the damage.

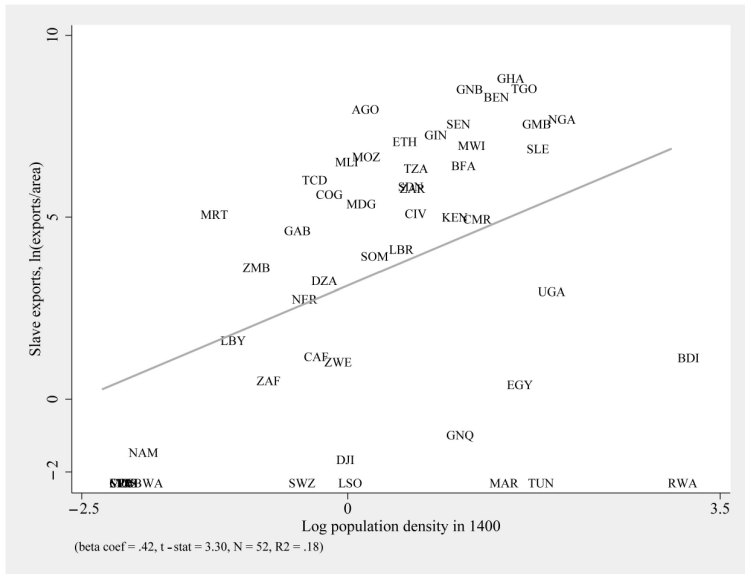


FIGURE IV
Relationship between Initial Population Density and Slave Exports

Comparing by distance to demand

- Second strategy: the distance from each country to the places where slave labor was demanded.
- Where demand sat was determined outside Africa. Sugar and tobacco in the West Indies and the southern United States. Gold and silver mines in **Brazil**. Salt mines in the northern Sahara, Arabia and Persia. Pearl divers in the Red Sea.
- The location of demand influenced the location of supply, and not the other way round.
- Four distances, one per trade: to the nine largest Atlantic importers (Virginia, Havana, Haiti, Kingston, Dominica, Martinique, Guyana, Salvador and Rio de Janeiro); to Mauritius and Muscat; to Algiers, Tunis, Tripoli, Benghazi and Cairo; to Massawa, Suakin and Djibouti.

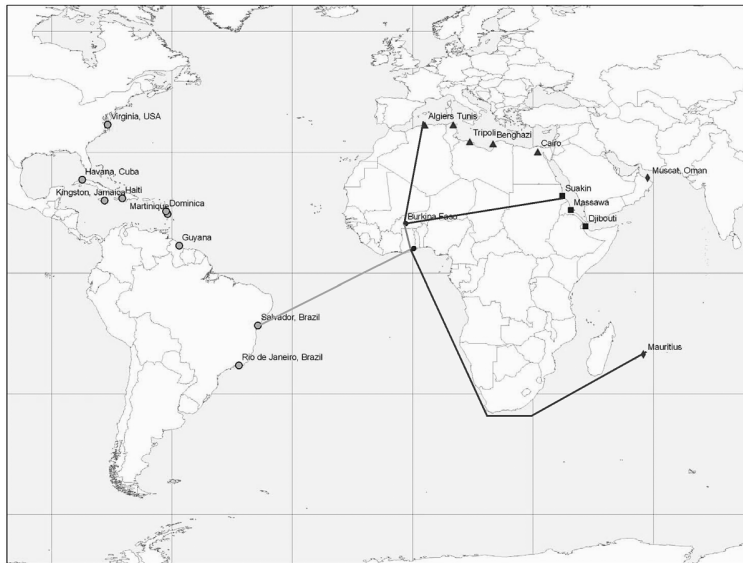


FIGURE V
Example Showing the Distance Instruments for Burkina Faso

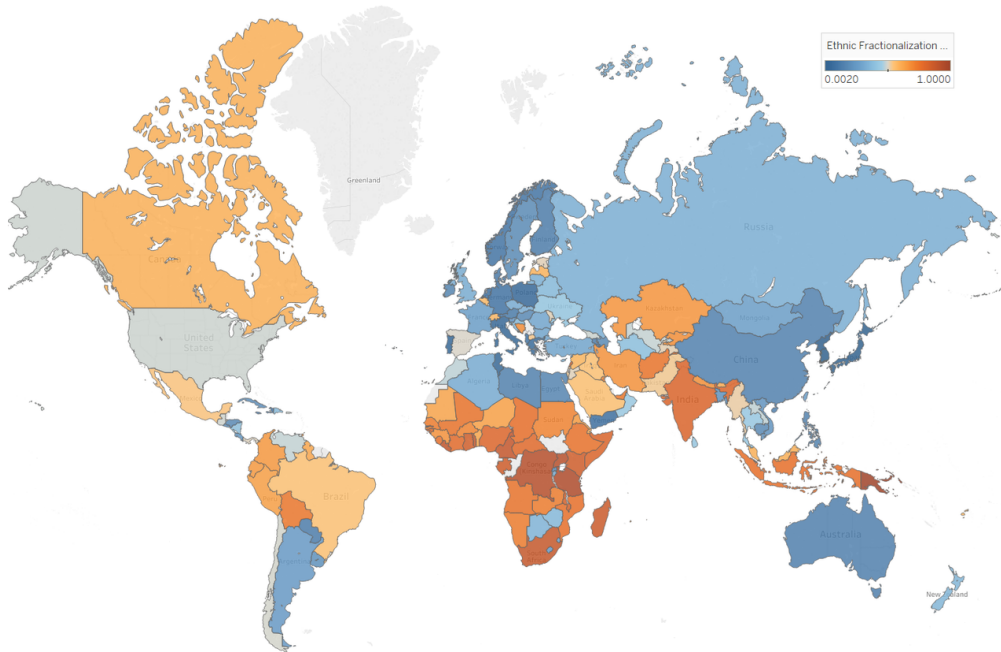
What the comparison shows

- Figure V draws the four distances for **Burkina Faso**.
- Within Africa, countries further from the slave markets are richer today. Being further from slave markets was good for growth.
- Outside Africa, the same distances bear no clear relationship to income.
- If the distances mattered for some other reason, that reason should show up outside Africa too.
- The damage implied by this comparison is *larger* than in the simple one, as expected when both undercounting and selection push the simple comparison toward zero.

Through which channels?

Channel 1: ethnic fractionalization

- Nunn calls this part preliminary and exploratory: 52 countries only.
- The slave trades weakened ties between villages, discouraging larger communities and broader ethnic identities.
- Countries that exported more slaves are **more ethnically fractionalized** today (measure from Alesina et al. 2003).
- Ethnic diversity matters for social cohesion, domestic institutions and policies, and the quality of government.
- It also reduces the provision of public goods: education, health facilities, access to water, transport infrastructure.



Channel 2: weak states

- Precolonial **state development**, from Gennaioli and Rainer (2006), built on Murdock (1967).
- The measure counts jurisdictional hierarchies beyond the local community, 0 to 4. Zero is a “stateless” society, four a “large state”.
- Countries that exported more slaves score lower on it.
- State antiquity is positively correlated with growth. Within Africa, ethnicities with centralized precolonial institutions provide more public goods today.

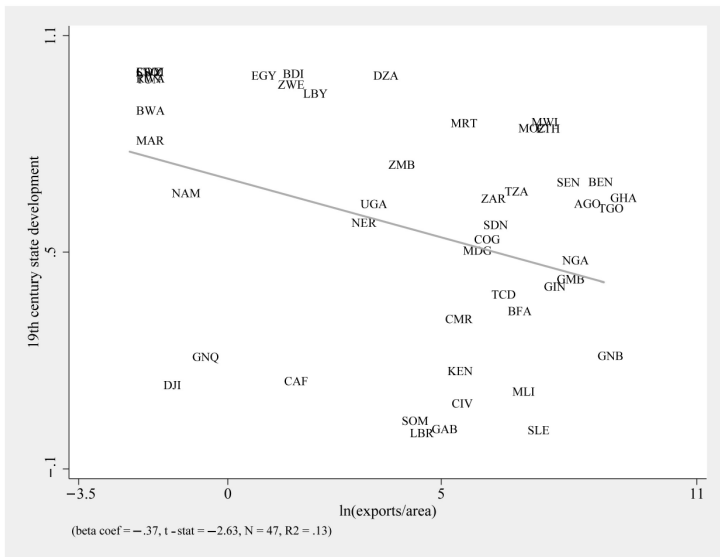


FIGURE VII
Relationship between Slave Exports and Nineteenth-Century State Development

Falling behind after independence

- Herbst: Africa's poor performance is postcolonial state failure, rooted in the instability of precolonial polities.
- Corollary: the effect of the slave trades should be felt most strongly *after* independence, when precolonial structures suddenly mattered more.
- Figure VIII follows average per capita GDP from 1950 to 2000 for the 26 countries with the lowest and the 26 with the highest slave exports.
- The low-export group was already richer in the early 1950s, when most countries were still under colonial rule.
- The gap **widens** over time, most pronounced after the late 1960s. In 1950 only four African countries were independent; by 1969, 42 of 52.

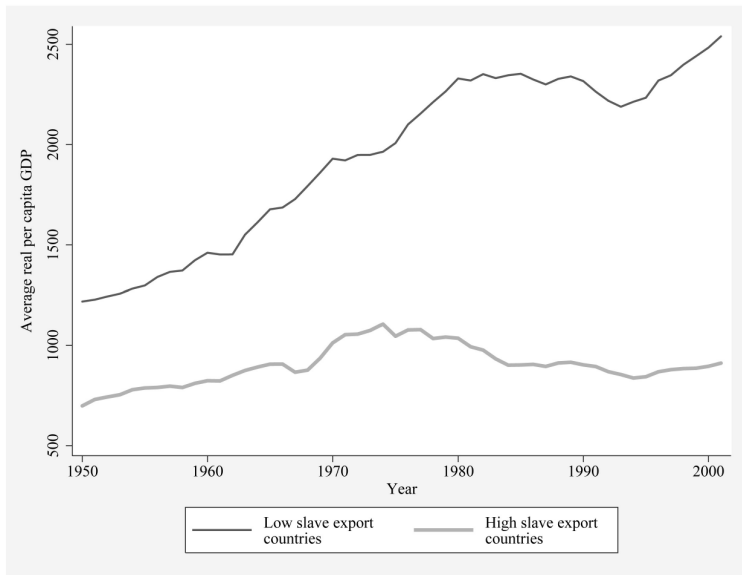


FIGURE VIII
Paths of Economic Development Since 1950

Conclusion: deep roots of underdevelopment

- A robust negative relationship between the number of slaves taken from a country and its subsequent economic development.
- Selection ran the wrong way for the spurious story, and the distance comparison confirms the result.
- Engerman and Sokoloff (1997, 2002) showed that the *use* of slaves in the New World produced institutions that were not conducive to growth.
- Nunn adds the **production** of slaves, through domestic warfare, raiding and kidnapping.
- The channels consistent with the data: ethnic fractionalization, and the underdevelopment of states.

A mechanism: trust

Nunn and Wantchekon (2011, AER)

From Nunn (2008) to Nunn and Wantchekon (2011)

- Nunn (2008) identifies a negative causal relationship between the slave trade and income today, but cannot establish the **causal mechanisms**.
- Here: one channel. Did the slave trade cause a culture of mistrust to develop within Africa?
- Early in the trade, nearly all slaves were taken in large-scale conflicts or raids, creating extreme insecurity outside the local community.
- As the trade progressed, individuals turned on others close to them, including friends and family, to kidnap, trick and sell them into slavery.

The feature that makes this trade different

- In most environments of conflict, insecurity comes from outside. Here individuals could partially protect themselves by **turning against others within their own community**.
- Trickery, local kidnappings and small scale violence bought guns and iron weapons.
- Slave merchants and raiders formed strategic alliances with groups inside villages and states in order to extract slaves.
- Hawthorne: “the Atlantic slave trade was insidious because its effects penetrated deep into the social fabric of the Upper Guinea Coast, beyond the level of the state and to the level of the village and household”

The Beafares of Guinea-Bissau

- Households located near the ports profited by tricking unsuspecting strangers and selling them to merchants.

“these Beafares are so smart, that if a yokel arrives from the interior, they pretend that they want to give him shelter, and they receive him into their homes. After a few days have passed, they persuade him that they have friends on the ships, and that they would like to take him and have a party. But when they go to the ships, they sell him. In this way they trick many yokels”

André Alvarez d’Almada (1984, p. 121), quoted in Hawthorne (2003)

How people entered slavery at the end of the trade

- In the 1840s the German missionary and linguist Sigismund Koelle recorded the manner of enslavement of **144 former slaves** living in Free Town, Sierra Leone.
- Kidnapping: just over 40 percent. Captured during wars: just under 25 percent.
- Sold by relatives or friends: almost 20 percent. One account is of a slave “enticed on board of a Portuguese vessel” by “a treacherous friend”.
- Through the judicial system: 16 percent. Accusations of witchcraft, theft, adultery or murder, with the accused and often the family sentenced to slavery.
- The Kabre of Northern Togo developed the custom of selling their own kin.

Culture as rules of thumb

- The authors take their notion of culture from evolutionary anthropology: decision-making **rules-of-thumb** for uncertain or complex environments (Boyd and Richerson).
- Where acquiring information is costly or imperfect, general beliefs about the right thing to do save the individual that cost.
- Which rules spread is settled by their relative payoffs.
- The hypothesis: where the slave trade made insecurity general, rules based on mistrust paid better than rules based on trust.
- East Germany under the Stasi: an individual's best option was often to become an informant, turning on those close to him or her.

Why it could still be visible a century later

- Cultural outcomes can have multiple equilibria, and a society can settle permanently in the low-trust one.
- Norms and institutions reinforce each other: less cooperation leads to weaker enforcement, which leads to worse behavior and still less trust.
- Or the shock has simply not dissipated yet. Germany was divided for 45 years, and the beliefs it created are estimated to take 20 to 40 years to converge back.
- The slave trade lasted well over 400 years and ended just over 100 years ago.

The data: the Afrobarometer

- The 2005 **Afrobarometer**: nationally representative surveys, interviews in local languages, random samples of 1,200 or 2,400 individuals of voting age per country.
- About 21,000 respondents in 17 sub-Saharan countries: Benin, Botswana, Ghana, Kenya, Lesotho, Madagascar, Malawi, Mali, Mozambique, Namibia, Nigeria, Senegal, South Africa, Tanzania, Uganda, Zambia, Zimbabwe.
- Not covered: West Central Africa (Cameroon, Gabon, DR Congo, Angola) and the countries inland of the Red Sea (Sudan, Ethiopia, Eritrea).

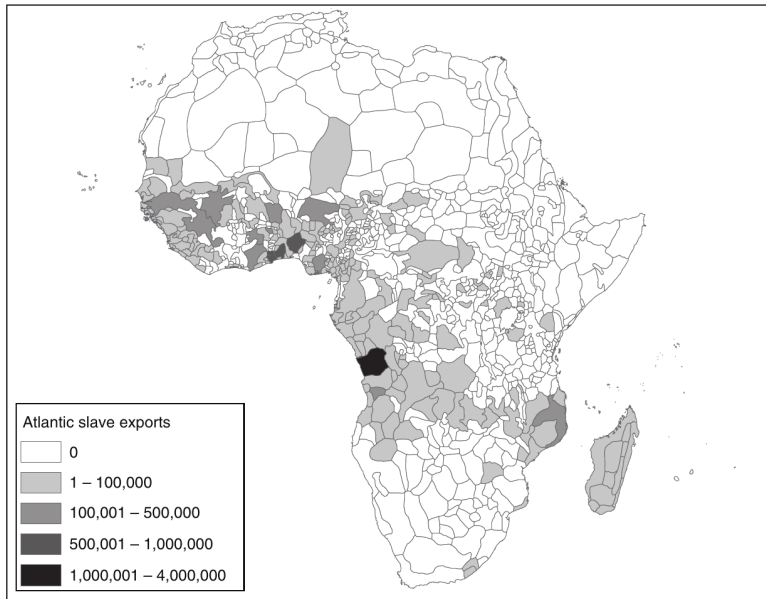
How trust is measured

- Five questions: trust in **relatives**, in **neighbors**, in the locally elected **government council**, in people of the same ethnic group, in people of other ethnic groups.
- Four answers, scored: not at all = 0, just a little = 1, somewhat = 2, a lot = 3.
- People report more trust in those closer to them than in those further away.
- Even about their own relatives, 7 percent say they do not trust them at all, and 18 percent only a little.

Linking trust today to the slave trade

- The country totals from Nunn (2008), broken down to the **ethnicity level** using the same ethnicity samples.
- Possible only for the transatlantic and Indian Ocean trades, the two with detailed enough ethnicity data. The transatlantic was by far the largest.
- Historical ethnic boundaries from Murdock (1959), whose classification is close to the one the Afrobarometer uses.
- Darker shades on the maps: more slaves shipped from that ethnic group between 1400 and 1900.

Panel A. Transatlantic slave trade



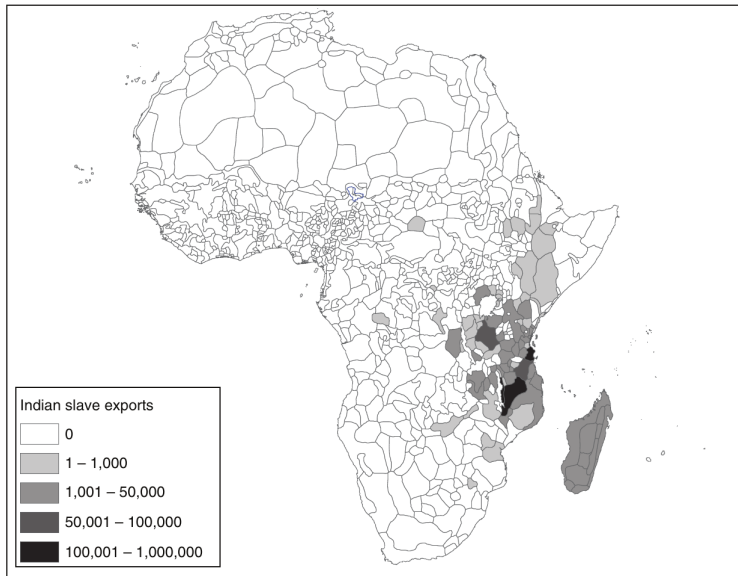


FIGURE 1

Reading the two maps

- Panel A: the transatlantic trade reached much of the continent. Not only West Africa and West-Central Africa, but also Eastern Africa and Madagascar.
- Panel B: the much smaller Indian Ocean trade was confined primarily to Eastern Africa.
- Both patterns match the qualitative evidence on where slaves in each trade came from.

The central result

- Individuals whose ancestors were heavily raided during the slave trade are **less trusting today**.
- It holds for all five measures: relatives, neighbors, the local council, coethnics, other ethnic groups.
- Slave exports and every other characteristic measured (age, gender, education, occupation, religion, living conditions, the ethnic composition of the district) together account for 5.4 percent of the variation in trust of neighbors. Of that, slave exports alone account for 16 to 27 percent.
- Consistent with the history: by the end of the trade, people were being sold by neighbors, friends and family members.

Does the slave trade cause the mistrust?

Alternative explanations

- Ethnic groups that were inherently less trusting may have been easier to raid, and may still be less trusting.
- Colonial rule followed the slave trade, from 1885 until independence. If it hit the same places, its effects could be credited to the slave trade.
- Three strategies.

Three strategies

- **Observables**: account for colonial rule and for precolonial characteristics of each group, including initial prosperity and political development. Little effect on the result.
- **Unobservables**: calculate how much stronger the unmeasured factors would have to be, relative to everything measured, to explain the result away. About four times stronger, which is implausible.
- **Distance from the coast**: before the transatlantic and Indian Ocean trades, Africans were not engaged in overseas external trade, so proximity to the ocean is unlikely to affect trust except through the slave trade.
- Distance to the Saharan trade routes and towns, European influence, and historical reliance on fishing are all accounted for separately.

Distance from the coast, inside and outside Africa

- Within Africa, groups whose ancestors lived further from the coast are more trusting today.
- The 2003 Asiabarometer, ten Asian countries (Japan, South Korea, China, Malaysia, Thailand, Vietnam, Myanmar, India, Sri Lanka, Uzbekistan): no relationship between distance from the coast and trust in local government.
- The 1990 World Values Survey (Chile, Norway, Sweden, Great Britain, Northern Ireland): no relationship either.
- Within that same survey, the one African country, **Nigeria**, shows the African pattern again. So the contrast is not an artifact of using different surveys.

Internal norms or external environment?

Two channels

- **Internal**: norms, beliefs and values, transmitted from parents to children.
- **External**: the slave trade caused a long-term deterioration of legal and political institutions, so weak enforcement permits poor behavior and the people around you really are less trustworthy.
- The two carry different implications, and the authors run three tests to weigh them.

Test 1: is the local council simply bad?

- Respondents with more heavily affected ancestors are more likely to disapprove of their local councilor, to report that more councilors are corrupt, and to feel that councilors do not listen.
- Accounting for those three perceptions cuts the relationship by just under 50 percent.
Over half remains.
- Adding objective measures of local government quality, whether the village has a school, a health clinic, sewage, piped water and electricity, changes little.

Test 2: are the neighbors themselves less trustworthy?

- For trust between groups, account for how heavily the slave trade hit the *other* ethnic groups living in the respondent's town, district or region.
- Those groups are indeed **trusted less** today, which is itself evidence that the slave trade damaged trustworthiness.
- Even so, the relationship with the respondent's own ancestry is barely affected.

Test 3: those who moved

- When individuals relocate, their beliefs and values move with them. Their external environment is left behind.
- Two measures: slaves taken from the respondent's **ancestral ethnic group**, and slaves taken from the place where the respondent lives today.
- Movers are 45 percent of the sample, and they are what makes the comparison possible.
- Both measures matter. The ancestral one is always at least *twice* the size of the location one.

Conclusion: the origins of mistrust

- Low levels of trust within Africa can be traced back to the transatlantic and Indian Ocean slave trades, and the evidence points to a causal relationship.
- Internal norms account for at least half of it. Test 1 puts them above 50 percent for trust in the local council; tests 2 and 3 put them at 85 to 100 percent for interpersonal trust.
- Trust matters for development, for trade, for political institutions and for how firms are managed. The contribution here is on **where the differences come from**.
- With Nunn (2008): the slave trade appears as a cause both of Africa's impoverishment and of its lower trust.